

Compression Therapy and Management of Venous Disease

Compression bandaging provides effective support during wound healing. A common method involves using a zinc paste-impregnated gauze (e.g., Unna boot) as the primary layer, followed by an outer layer of **Coban™** applied at full stretch to ensure consistent compression (see Figure 175-8). This approach delivers sustained, graduated pressure that aids venous return and reduces edema.

After ulcer healing, long-term dynamic support is essential to prevent recurrence. Patients should be fitted with graduated compression stockings that provide **30–40 mmHg** of pressure at the ankle. These stockings must be properly sized and worn daily for life to maintain venous health.

> **Important:** Compression stockings should not be applied to acutely edematous, inflamed, or tender limbs. Instead, use multilayer compression bandaging until edema and inflammation resolve, then transition to custom-fitted stockings.

In selected cases, correction of underlying venous incompetence can be achieved through **sclerotherapy** or **endovenous ablation** (e.g., laser or radiofrequency), particularly for incompetent perforator veins. These interventions address the hemodynamic abnormalities contributing to venous ulceration (see Chap. 250).

Prevention

Preventing **venous thrombosis** is key to avoiding chronic venous insufficiency. Given the genetic component of many thrombophilias, identifying high-risk individuals before a thrombotic event allows for targeted prophylaxis. In hospitalized patients or those undergoing elective surgery, appropriate antithrombotic measures significantly reduce the incidence of post-thrombotic syndrome.

For patients who have already experienced deep vein thrombosis (DVT), **elastic compression stockings** remain the only evidence-based intervention proven to reduce the risk of post-thrombotic complications.

Other preventive strategies include: - Use of **supportive compression stockings during pregnancy**, despite limited evidence, due to increased risk of valvular insufficiency. - Wearing compression stockings during prolonged immobility, such as long-distance air travel or sedentary occupations.

Other Venous Diseases

Deep Vein Thrombosis (DVT)

Epidemiology

DVT is a major cause of morbidity and mortality, primarily due to its potential to lead to **pulmonary embolism (PE)** and **chronic venous insufficiency**.

Etiology and Pathogenesis

DVT arises from a combination of **venous stasis, endothelial injury, and hypercoagulability** (Virchow's triad). Risk factors include surgery, immobilization, malignancy, pregnancy, oral contraceptives, and inherited thrombophilias.

Clinical Findings

Clinical diagnosis of DVT is unreliable. Common signs include: - Limb swelling - Pain or tenderness - Erythema or discoloration - Warmth - Superficial venous dilation

However, fewer than one-third of patients suspected of DVT based on symptoms actually have the condition.

Laboratory Tests

- **D-dimer** testing may help rule out DVT in low-risk patients.
- **Compression ultrasonography** is the standard diagnostic tool for confirming DVT.

Treatment

Anticoagulation is the cornerstone of DVT management. Treatment decisions should be made in consultation with appropriate specialists. Primary prevention strategies align with those used for peripheral vascular disease, including mechanical and pharmacologic prophylaxis in high-risk settings.

Superficial Venous Thrombosis and Phlebitis

Epidemiology

Superficial venous thrombosis (SVT) commonly occurs in varicose veins but can also be associated with systemic conditions such as thrombophilia, malignancy, or autoimmune disorders.

Etiology and Pathogenesis

SVT may result from venous stasis, trauma, catheter insertion, or inflammation. It can progress to DVT, especially when the clot is near the saphenofemoral junction.

Clinical Findings

Superficial phlebitis presents with: - Localized pain - Tenderness - Induration - Erythema - Palpable venous cord

A persistent cord upon limb elevation suggests thrombus formation rather than pure inflammation.

Eponymous forms include: - **Mondor disease**: Thrombophlebitis of the chest wall or dorsal penile vein. - Lower extremity SVT: Often involves the great saphenous vein.

Laboratory Tests

Due to the risk of concurrent DVT, some experts recommend **compression ultrasound** in all patients with lower extremity SVT, particularly if the clot is within 3 cm of the saphenofemoral junction. However, this approach lacks strong trial evidence and should be guided by clinical judgment.

Treatment

Treatment depends on location and risk: - **Low-risk SVT** (distal, small veins): NSAIDs, warm compresses, and local care. - **High-risk SVT** (proximal, near deep system): Consider anticoagulation (e.g., low molecular weight heparin or fondaparinux) to prevent extension into deep veins.

Lymphedema

Lymphedema results from chronic impairment of lymphatic drainage, leading to accumulation of protein-rich fluid in the interstitial space. It typically manifests as non-pitting edema, most commonly in the limbs.

The pathophysiology involves disruption of normal fluid dynamics: capillary filtration exceeds lymphatic clearance, causing prolonged edema and eventual tissue fibrosis.

Lymphedema may be: - **Primary** (congenital or developmental abnormalities) - **Secondary** (due to surgery, radiation, infection like filariasis, or chronic venous disease)

Early diagnosis and intervention are critical to prevent progression and complications such as recurrent cellulitis and skin changes.